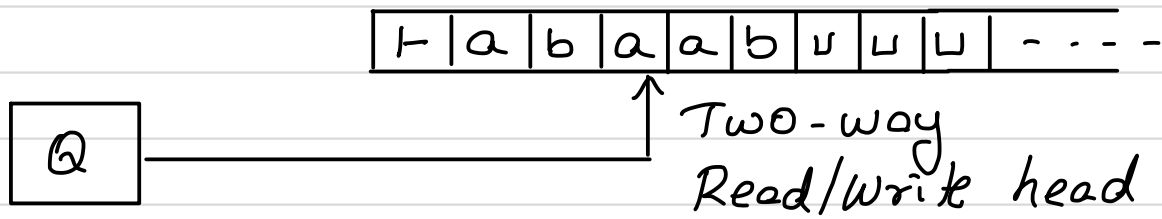


Deterministic Turing Machine



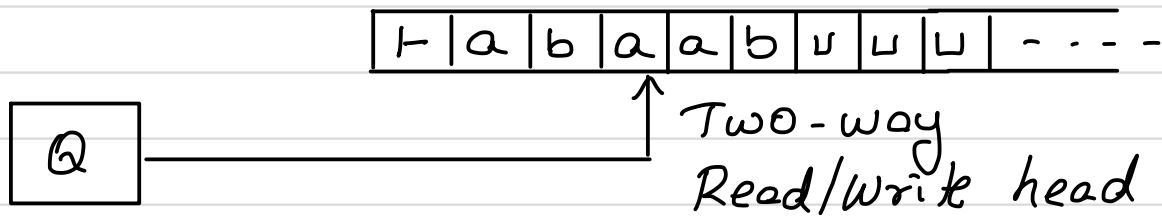
Finite set of States + Semi-infinite tape

- infinite to the right.
- Delimited on the left with an end marker $\sqcup$

Head - Can move left or right over the tape, can read and write symbols on the tape.

Infinitely many cells to the right of the input string contains a special blank symbol $\sqcup$

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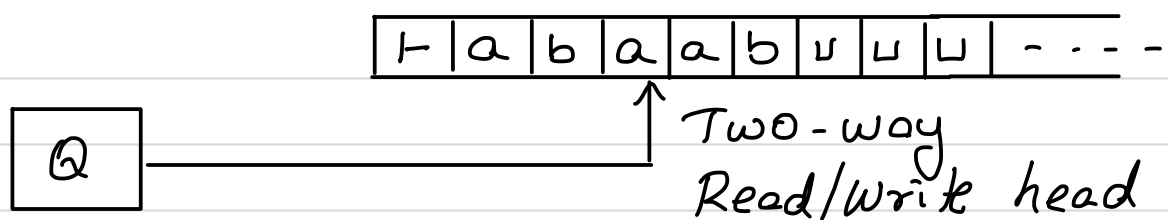
- infinite to the right
- Delimited on the left with an end marker $\vdash$

Head - Can move left or right over the tape, can read and write symbols on the tape.

Infinitely many cells to the right of the input string contains a special blank symbol $\sqcup$

Operation of a Turing Machine.

- Start in the start state & with head scanning the left end marker $\vdash$
- Each step, read the symbol on the tape under its head
- Depending on the symbol and current state, write a new symbol on that tape cell.
- Move the head left or right and change to a new state
- Accept input by entering a special accept state.
- Reject input by entering a special reject state.



Formal definition of a Turing machine

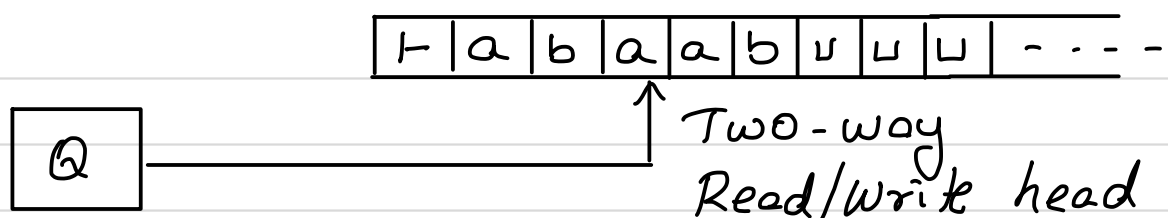
$$M = (Q, \Sigma, \Gamma, \vdash, \sqcup, \delta, \delta', t, \tau)$$

Σ : finite input alphabet

Γ : finite tape alphabet [$\Sigma \subseteq \Gamma$]

$\sqcup \in \Gamma - \Sigma$: blank symbol

$\vdash \in \Gamma - \Sigma$: left end marker



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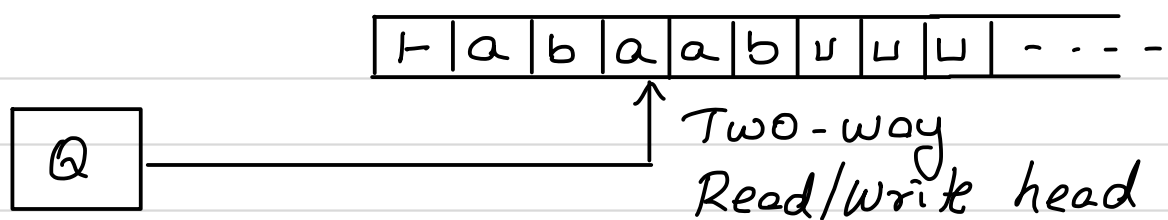
$s \in Q$: start state.

$t \in Q$: Accept state, $\gamma \in Q$: Reject state

$$\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$$

$$\delta(p, a) = (q, b, d)$$

$$\textcolor{red}{\hookrightarrow d \in \{L, R\}}$$



Formal definition of a Turing machine

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$$\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$$

$$\delta(p, a) = (q, b, d)$$

$$\text{ } \xrightarrow{\text{red}} d \in \{L, R\}$$

Assumptions:

- $\forall p \in Q, \exists q \in Q$ such that $\delta(p, \vdash) = \delta(q, \vdash, R)$

- $\forall b \in \Gamma, \exists c, c' \in \Gamma$ and $\exists d, d' \in \{L, R\}$ such that

$$\left. \begin{array}{l} \delta(t, b) = (t, c, d) \\ \delta(r, b) = (r, c', d') \end{array} \right\} \text{red arrow}$$

Once m enters accept/reject state it never leaves it.

not a CFL

Example. A TTM that accepts $\{a^n b^n c^n \mid n \geq 0\}$

1. Start in q_0 , scans to the right over the input once to ensure the input is of the form $a^* b^* c^*$
 - writes to the tape the same symbol it reads
 - writes an end marker at the end of the input.
2. Scan the input over rounds to match one a , b and c and erase these symbols.

Example: $a^4 b^4 c^4$

	a	a	a	a	b	b	b	b	c	c	c	c	□	□	- - -
--	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-------

Configuration of a Turing machine

Tape of M consists of a string $y\sqcup^\omega$ where $y \in \Gamma^*$.

A configuration is $\in Q \times \{y\sqcup^\omega \mid y \in \Gamma^*\} \times \mathbb{N}$
 (p, z, n)
 $Q \leftarrow$ $\rightarrow$ Current position of head on tape
 $\rightarrow$ Current content of the tape.

$\underbrace{\hspace{1.5cm}}_{\text{infinite string with a finite presentation.}}$

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 (p, z, n)
 $Q \leftarrow$ state
 $n \geq 0$ current position of head on tape
current content of the tape.
 $\underbrace{\{y \sqcup^\omega \mid y \in \Gamma^*\}}_{\text{infinite string with a finite presentation.}}$

Start configuration - $(q, \vdash x \sqcup^\omega, 0)$
input string

Next configuration relation $\xrightarrow{\frac{1}{m}}$

For an infinite string z , let z_n be the n^{th} symbol in z

$S_b^n(z)$ - string obtained by substituting b for z_n .

Example $S_b^3(\vdash a b a a \dots) = \vdash a b b a \dots$

$$(p, z, n) \xrightarrow{\frac{1}{m}} \begin{cases} (q, S_b^n(z), n-1) & \text{if } \delta(p, z_n) = (q, b, L) \\ (q, S_b^n(z), n+1) & \text{if } \delta(p, z_n) = (q, b, R) \end{cases}$$

$\alpha \xrightarrow{*} \beta$: Reflexive transitive closure of $\xrightarrow{\frac{1}{m}}$

M accepts $x \in \Sigma^*$ if $(s, \vdash x \sqcup^w, 0) \xrightarrow{*}_M (t, y, n)$

$$L(M) = \{x \in \Sigma^* \mid M \text{ accepts } x\}$$

M rejects $x \in \Sigma^*$ if $(s, \vdash x \sqcup^w, 0) \xrightarrow{*}_M (r, y, n)$

M **halts** on input $x \in \Sigma^*$ if it either accepts x or rejects x .

M **loops** on input $x \in \Sigma^*$ if M neither accepts nor rejects x .

M accepts $x \in \Sigma^*$ if $(s, \vdash x \sqcup^w, 0) \xrightarrow{*}_M (t, y, n)$

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M **halts** on input $x \in \Sigma^*$ if it either accepts x or rejects x .

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Total TM: A Turing machine that halts on all inputs is called a total TM.

$A \subseteq \Sigma^*$ is **recursive** if $A = L(M)$ for a total TM M .

$A \subseteq \Sigma^*$ is **recursively enumerable (r.e.)** if $A = L(M)$ for some TM M (need not be a total TM).

Ex. $A = \{a^n b^n c^n \mid n \geq 0\}$ is recursive.